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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 11/10/2025 |
| **PI Name:** | Shelhamer, Mark Sc.D. | | |
| **Project Title:** | Assessment of Otolith Function and Asymmetry as a Corollary to Critical Sensorimotor Performance in Missions of Various Durations | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | Yes |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 21205-1832 | **Congressional District:** | 7 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C |
| **Start Date:** | 02/07/2019 | **End Date:** | 02/07/2033 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 3 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Brocato, Becky | **Contact Phone:** | |
| **Contact Email:** | becky.brocato@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 2/7/2033 per L. Barnes-Moten/JSC and NSSC information (Ed., 5/18/21)<br>NOTE: End date changed to 6/6/2021 per NSSC information (Ed., 11/9/20) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Schubert, Michael Ph.D. ( Johns Hopkins University )<br>Kheradmand, Amir M.D. ( Johns Hopkins University School of Medicine ) | | |
| **Grant/Contract No.:** | 80NSSC19K0487 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | We propose a simple and rapid measure of otolith asymmetry -- a basic low-level neurovestibular function that is altered in spaceflight. Our assessment apparatus and procedure was developed with previous Human Research Program (HRP) support, and has undergone validation testing and is currently in use in an unrelated Department of Defense (DoD) study. It provides a measure of binocular alignment (vertical and torsional), which is a reflection of underlying otolith asymmetry. While this measure might be important in itself (since it reflects a form of central neural compensation that might be challenged in spaceflight), it could also provide a simple alternative to locomotion and posture testing if it is found to correlate with those measures. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | A simple and rapid assessment of neural/neurovestibular function, as described here, will be a useful in-flight procedure to track changes over the course of a mission. Given the widespread connection of neural circuits and functions with other body functions, it is not inconceivable that neurovestibular effects as assessed here will likewise have implications for a wide variety of neural, cognitive, and performance aspects of crews in flight. Our testing provides an opportunity to assess a low-level neural system that is not under volitional control, is responsive to g-level alterations, and might be an indicator of broader and more functionally relevant effects. The technology and procedures for this project are being used in DoD projects for the assessment of vestibular function in military personnel. |
| **Task Progress:** | Poster presentation was made at the NASA Human Research Program (HRP) Investigators Workshop in February 2025. In the fall of 2021, we started a set of experiments in the lab, to provide further baseline data by which to evaluate the upcoming flight data from this project. In these trials, subjects are evaluated for ocular alignment (OA) while standing on a hard floor, standing on foam, while seated, and while making head movements. This will help us evaluate the impact of postural demands on ocular alignment; we conjecture that the heightened demand while standing will call for a more precise compensation for any latent otolith asymmetry, causing a reduction in misalignment in that condition compared to seated. Experiments were on hold during changes in the student investigator team. At this time approximately 25 subjects have been tested, and data analysis is continuing. Indications so far are that there is no clear relationship of otolith asymmetry to posture or locomotion parameters, although a recent finding suggests that dispersion of body motion about a static standing position might be related to larger VAN/TAN values. Similar analyses will be applied to a set of data obtained from a normative population at Space Center Houston.<br><br>This project also includes a component dedicated to data-integration of the other Complement of Integrated Protocols for Human Exploration Research (CIPHER) projects. While awaiting CIPHER flight data, we have been performing several preparatory activities so that we can be better prepared to organize and analyze that data:<br><br>1. A manuscript describing the scientific background and justification for CIPHER has been prepared by the PI. Significant rewriting of this manuscript is now necessary, given the significant re-scoping of CIPHER in 2025. Currently, this is awaiting further input from CIPHER leadership.<br><br>2. Discussions continue on how to acquire, archive, and curate the data that are expected to be provided to the PI by the other CIPHER investigators. The decision was made, jointly with CIPHER leadership, that NLSP is the right platform, provided a means for incorporating medical and metadata is developed, and that the PI can have access to a work area on the platform.<br><br>3. Through discussions with Anand Narayanan, we identified a set of data in OSDR (Rodent Research 9) that were analyzed for relations between bone and ocular measures, as an initial investigation of cross-disciplinary effects of spaceflight. Since this project analyzed significant data before CIPHER, it will provide guidance as to relevant cross-discipline analysis methods. This was assisted by data-science student Xiangyu Zhang and her faculty advisor Brian Caffo. This work was accepted for presentation at the 2025 ASGSR conference and will be prepared for publication as well.<br><br>4. Ultimately, multivariate statistics and machine learning approaches will be applied to the large CIPHER dataset to identify correlations between measures. In the meantime, we have conducted an initial analysis of one of the commercial spaceflight datasets to which we have access, taking a very general approach to how various physiological measures covary or diverge while in flight relative to the preflight baseline. This analysis makes use of entropy measures of various kinds. In particular, we have been using Functional Entropy (FE), which quantifies “discordance” or “misalignment” between the various physiological parameters, relative to preflight baselines. Our initial use of this measure is very ad hoc, making use of the available physiological measures, with sporadic sampling and uncertain measurement reliability. Nevertheless, the first results are promising, as they show a dramatic spike in FE for all subjects at the time of launch and orbital entry, and substantial ongoing peaks in FE during the course of the mission in those crew who had SMS. Those same subjects also exhibited other forms of “discomfort” during the flight, as assessed by a member of the investigator team (who was naïve to SMS incidence) viewing publicly available cabin video. This indicates that entropy measures might capture essential features of “good” and “bad” spaceflight adapters, which raises the possibility of flagging undesired deficient adaptation and intervening appropriately.<br><br>5. VAN/TAN preflight data have been obtained for the initial set of subjects. These will be compared to ocular skew measures taken in other parts of the VH project, to assess similarities between the two ways of making this assessment (the other VH measure uses a head-mounted video system to measure eye positions directly). We will also examine VAN/TAN for correlation with SMS prevalence. A related TRISH study in our lab (EXP0008, ended 10/31/25) shows promise in predicting SMS susceptibility based on the pattern of preflight VAN/TAN values, and we will examine CIPHER data with this in mind. An initial analysis of VAN trial-by-trial data (22 trials in each case) for two subjects reveals an alternating up-down pattern that is more apparent in one subject and which may have relevance to the ability to learn from one trial to the next, which may in turn confer an advantage of adaptability which relates to the lack of SMS. This finding is new, and the analysis is in its very early stages, yet to be confirmed or refuted with SMS information. |
| **Bibliography Type:** | Description: (Last Updated: 01/12/2026) |

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| **Papers from Meeting Proceedings** | Zhang X, Narayanan A, Shelhamer M, Caffo B. "Multisystem physiological adaptations caused from spaceflight: A variational Bayesian CCA approach." 2025 American Society for Gravitational and Space Research (ASGSR) Annual Meeting, Phoenix, AZ, December 3-6, 2025.<br>2025 American Society for Gravitational and Space Research (ASGSR) Annual Meeting, Phoenix, AZ, December 3-6, 2025. , Dec-2025 |